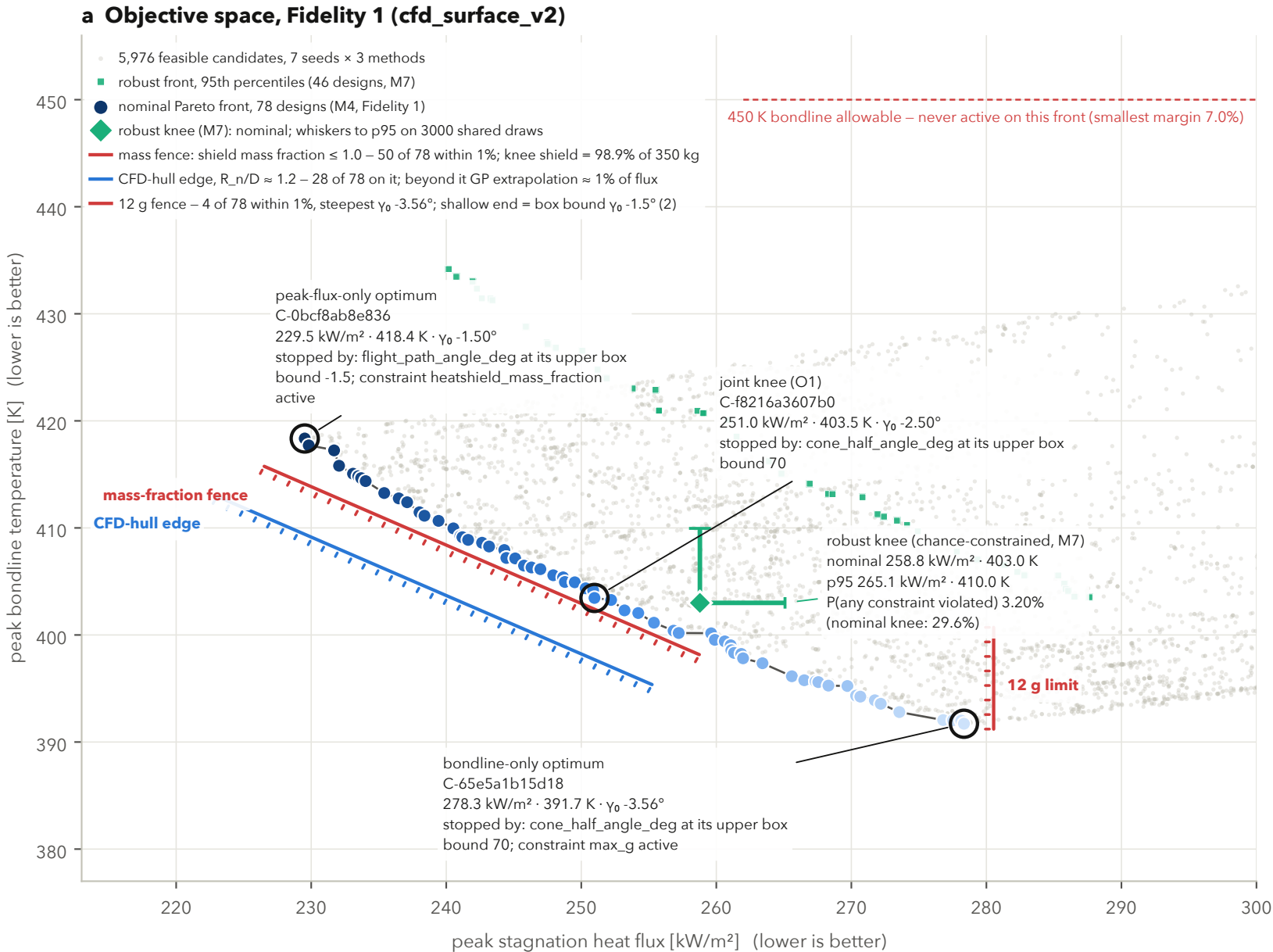
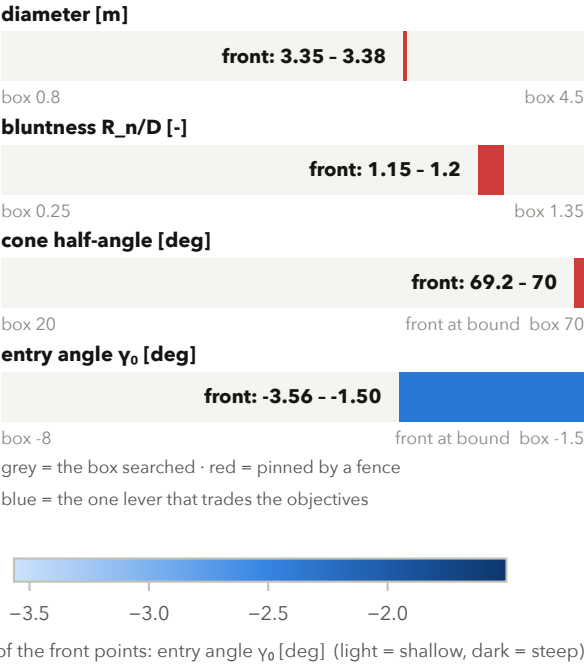


3 What the optimiser found: an entry-angle curve drawn at a fenced geometry

The 78-design front trades the two objectives through flight-path angle only.
Diameter, bluntness and cone angle are pinned by a mass-fraction fence, the edge of where CFD was run, and a box bound.



b What the front varies, against its box



knee against the peak-flux-only optimum

-14.9 K

bondline, for +21.5 kW/m² of peak flux

whole front: 26.6 K and 48.8 kW/m² of span.

Nominal knee under uncertainty: p95 410.4 K.

An absolute difference inside a placeholder stack, not a margin against a qualified material.

Front and cloud: results/M4/M4-OPT-20260920T205252Z/candidates.csv (front recomputed as the non-dominated feasible set; size asserted equal to summary.json). Named designs, fences, box audit: results/M4/M4-OPT-20260920T205252Z/summary.json (git f5a55b6, dirty).
Robust knee: results/M7/M7-LFL-20260921T011854Z/likeforlike.json, propagated through the 3000 nested draws of M7-UQ-20260920T233048Z; robust front: results/M7/M7-ROBUST-20260920T211216Z/robust_front.csv (32-draw common-random-number inner sample). Hull count: reports/milestones/M4_pareto_optimisation.md §11.
Fidelity 1 = a CFD-derived drag surface, not a validated one. Model predicts, within tested assumptions; 'optimum' names a design on this front, not a located capsule shape.